Information Gain & Decision Tree Notes

Outlook	Temperature	Humidity	Windy	Rain
Sunny	Hot	high	weak	No
Sunny	mild	high	strong	No
Cloudy	Hot	high	weak	Yes
Rainy	mild	high	weak	Yes
Rainy	cool	Normal	weak	Yes
Rainy	cool	normal	strong	No
Cloudy	mild	normal	strong	Yes
Sunny	cool	normal	weak	No

1. Entropy of the class from the data set:

Total records = 8

Yes = 4

No = 4

Entropy formula: $H(S) = \sum -p \log_2(p) = -p \log_2(p)$

$H(S) = -(4/8 \log_2(4/8) - 4/8 \log_2(4/8))$

$= -(0.5 \log_2(0.5) - 0.5 \log_2(0.5))$

class = 1

Now, prime attribute:

1. Outlook:

Entropy

	Yes	No	
Sunny	$-(0/3 \log_2(0/3)$	$+ 3/3 \log_2(3/3)) = 0$	
Cloudy	$-(2/2 \log_2(2/2)$	$+ (0/2 \log_2(0/2)) = 0$	
Rainy	$-(2/3 \log_2(2/3)$	$+ (1/3 \log_2 1/3)) = 0.918$	

Weighted:

$3/8(0) + 2/8(0) + 3/8(0.918)$

$= 0.344$

info gain = 1 - 0.344
= **0.655**

ii. Temperature:

	Y	N
Entropy		
Hot	$-(1/2 \log_2(1/2) + (1/2 \log_2(1/2)) = 1$	
mild	$-(2/3 \log_2(2/3) + 1/3 \log_2(1/3)) = 0.918$	
cool	$-(1/3 \log_2(1/3) + (2/3 \log_2(2/3) = 0.918$	

weighted

$2/8 (1) + 3/8 (0.918) + 3/8 (0.918)$

= 0.939

info gain = 1 - 0.939

= **0.061**

Humidity:

	Y	N
Entropy		
High	$-(2/4 \log_2(2/4) + (2/4 \log_2(2/4)) = 1$	
Normal	$-(2/4 \log_2(2/4) + 2/4 \log_2(2/4) = 1$	

weighted:

$4/8 (1) + 4/8 (1)$

= 1

info gain = 1 - 1

= **0**

~~Rainy~~ windy:

	Y	N
Entropy		
weak	$-(3/5 \log_2(3/5) + 2/5 \log_2(2/5) = 0.971$	
strong	$-(1/3 \log_2(1/3) + 2/3 \log_2(2/3) = 0.918$	

weighted. $3/8 (0.971) + 3/8 (0.918) = 0.951$

info gain = 1 - 0.951 = **0.049**

outlook is the highest attribute. Select it the root node.